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IOTA TANGLE 2.0: AN OVERVIEW

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Abstract. The IOTA Tangle cryptocurrency represents the next evolution in distributed ledger technology, designed to facilitate transactions in the Internet of Things applications. Compared to traditional blockchain architectures, the IOTA presents attractive characteristics, including high scalability, zero-fee transactions, higher achievable transactions per second, and lower energy consumption. The consensus mechanism in IOTA 1.0 and 1.5 relied on a centralized coordinator solution for transaction validation. IOTA 2.0, designed in 2020, is the first fully decentralized network that supports smart contracts. This paper provides a comprehensive overview of the IOTA Tangle 2.0.

INTRODUCTION

The Internet of Things (IoT) is a network of physical objects embedded with sensors, software, and network connectivity to enable them to collect and exchange data (Nguyen et al., 2023). The number of connected IoT devices worldwide is overgrowing and is expected to reach 29.42 billion by 2030, up from 15.14 billion in 2023 (Vailshery, 2022).

IoT-related applications such as smart homes, smart cities, smart transportation, smart manufacturing, smart healthcare, wearable devices, and smart agriculture positively affect people's lifestyles. However, IoT security is a major challenge that must be addressed to realize IoT's potential benefits fully (Jahangeer et al., 2023; McLaughlin, 2023).

Blockchain (BC) is a transparent, immutable, and decentralized ledger technology that has the potential to improve the security of IoT devices and systems significantly (Yousra et al., 2023). However, some challenges still need to be addressed before BC can be widely adopted for IoT security (Maleh et al., 2022). These challenges include scalability constraints, energy consumption, transaction fees, a limited number of transactions

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per second, and the need for standards for BC-based IoT security solutions (Abed et al., 2023; Conti et al., 2022; Lamriji et al., 2023b; Tyagi et al., 2023).

In 2015, IOTA cryptocurrency was designed to cater to the needs of IoT networks and applications. The IOTA Tangle is based on a directed acyclic graph (DAG) structure, which has high scalability, no-fee transactions, achievable higher transactions per second, and energy saving (Silvano & Marcelino, 2020).

The IOTA project has undergone significant developments and improvements since its inception. In IOTA 1.0 and IOTA 1.5, the consensus mechanism (CM) relied on a centralized coordinator solution for transaction validation and security. IOTA 2.0 removed this centralized coordinator and replaced it with a decentralized CM that allows nodes in the network to collectively validate transactions and achieve consensus without relying on a single entity. IOTA 1.0 signs transactions using the Winternitz One-Time signature (W-OTS). To improve the security and robustness of the network's transaction validation process, IOTA 1.5 and 2.0 switched to the Ed25519 signature scheme (Conti et al., 2022).

This work aims to provide an overview of the IOTA Tangle 2.0. First, it presents the architecture, the signing scheme, consensus mechanisms, and smart contracts of IOTA 2.0. Then, it provides an implementation of IOTA using Python.

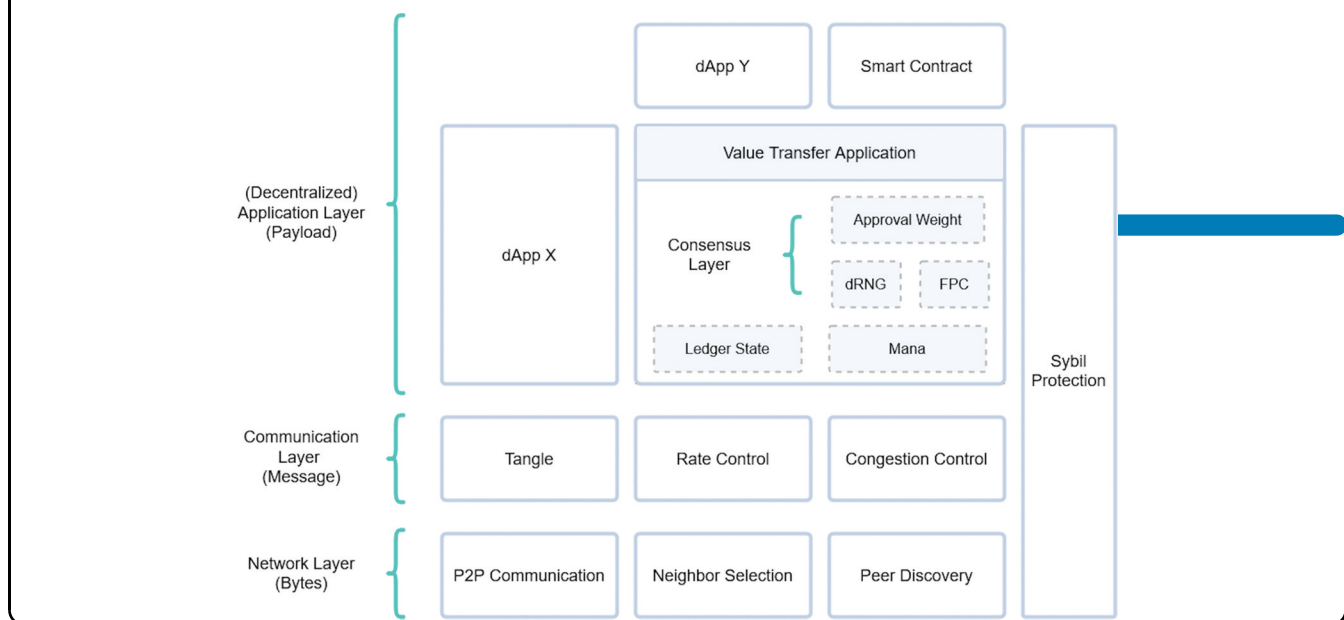
OVERVIEW OF THE IOTA 2.0 PROTOCOL

IOTATangle was designed to address some of the challenges associated with traditional BC systems, particularly in IoT applications. These challenges include scalability, cost, and latency (Conti et al., 2022). The IOTA protocol has evolved over the years through three major versions: 1.0 in 2015, 1.5 in 2017, and 2.0 in 2020. IOTA Tangle 2.0 has many features, including:

- **High scalability:** IOTA 2.0 is designed to support a high throughput of transactions per second (TPS). In contrast, the number of TPS is 7 for BC Bitcoin and 15 for BC Ethereum. As IOTA transactions increase, the network becomes more robust and leads to faster confirmations.
- **Efficiency:** IOTA 2.0 is very efficient, with no fees and no need for miners.
- **Decentralized consensus:** IOTA Tangle 2.0 uses a decentralized consensus mechanism called Curl-P, which makes it more secure and resistant to attack.
- **Smart contracts:** IOTA 2.0 supports SCs, which can be used to create decentralized apps.
- **Interoperability:** IOTA Tangle 2.0 is interoperable with other BCs, which makes it possible to transfer value between different networks.

Architecture of IOTA 2.0

The architecture of the IOTA 2.0 protocol, as presented in Figure 1, consists of multiple layers that collaboratively create

Figure 1 *Layers of the IOTA 2.0 protocol.*

a robust and high-performing distributed ledger system (Zhang, 2022). These layers encompass:

- **Network layer** : It focuses on the Tangle node functionality and manages the byte-level operations, including the P2P exchanges between nodes.
- **Communication layer** : Is responsible for handling IOTA messages. This involves essential tasks such as tip selection approvals, rate control mechanisms, and management of the Tangle ledger.
- **Application layer** : It takes charge of executing message payloads. For instance, in the case of a value transaction, it facilitates consensus execution, fund transfers, and reputation (mana) transfer processes.

Signature Scheme

IOTA 2.0 signs transactions using the Ed25519 signature scheme, replacing the W-OTS scheme used in IOTA 1.0. Ed25519 is a variant of the EdDSA scheme designed on the twisted Edwards curve to offer good performance and robust security properties (Bernstein et al., 2012; Conti et al., 2022). We implemented Ed25519 and W-OTS schemes using Python on a restrained device, a Raspberry Pi 1 [ARM 1 core CPU: 700 MHz and RAM: 512 MB]. The Ed25519 achieves the best performance when using No Adjacent Form to calculate scalar multiplication, the extended Euclid algorithm to calculate modular multiplicative inverse, the Shamir method for signing verification, and inverted coordinates

Table 1 Comparison of Ed25519 and W-OTS digital signature schemes.

Feature	Ed25519	W-OTS
Key size	32 bytes	256 bytes
Signature size	64 bytes	1300–3900 bytes
Key generation time	0.138 s	0.696 s
Signing time	0.149 s	0.371 s
Verification time	0.207 s	0.258 s
Security level	128 bits	112 bits

(Lamrij et al., 2022, 2023a). Table 1 gives some features of the Ed25519 signing scheme.

Ed25519 curve defined over a finite field F_p which consists of the points satisfying the equation:

$$(E_{Ed25519}) : -x^2 + y^2 \equiv 1 + d*x^2*y^2 \pmod{p} \quad (1)$$

Where,

- $d = -\frac{121665}{121666}$
- $p = 2^{255} - 19$ is the prime number that defines the F_p .
- The base point: $G = 15112221349535400772501151409588531511454012693069428394003475163141307993866256225615783033603165251855960).41857206046113283949847762202, 463168356949264781$
- The order q of $q : q = 2^{252} + 27742317777372353535851937790883648493$.
- The cofactor is 8.

The three constituent algorithms of Ed25519 are key generation (Algorithm 1), signature generation (Algorithm 2), and verification (Algorithm 3).

Algorithm 1 Ed25519 key generation

Require: The prime p , $E_{Ed25519}$ coefficient (d), the generator G , and the prime q .

- 1) Choose a random integer $k \in \{0,1\}^{256}$.
- 2) Compute $h = \text{SHA512}(k)$.
- 3) Drive x from the second 256 bits of h .
- 4) Compute $X = xG$.

Ensure: The public key is $k_{\text{pub}} = (p, d, q, G, X)$, and the private key is $K_{\text{pr}} = (k)$.

Algorithm 2 Ed25519 signature generation

Require: K_{pub} , K_{pr} , and a message m .

- 1) Recompute h and x .
- 2) Drive y from the first 256 bits of h .
- 3) Compute $r = \text{SHA512}(y || m) \pmod{q}$.
- 4) Compute $R = rG$.
- 5) Compute $S = (r + \text{SHA512}(R || X || m) * x) \pmod{q}$.

Ensure: (R, S) .

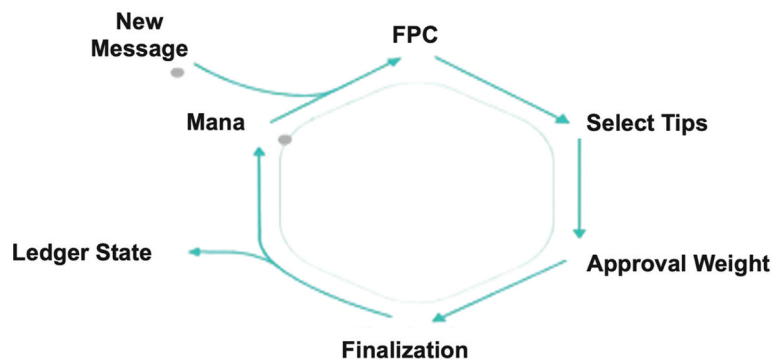
Algorithm 3 Ed25519 verification**Require:** K_{pub} , (R, S) , and the message m .

- 1) Compute $z = 8\text{SHA512}(R||X||m)(\text{mod } q)$.
- 2) Compute $P_1 = 8SG$.
- 3) Compute $P_2 = 8R + zX$.
- 4) **if** $P_1 = P_2$ **then return** Signature Valid
- 5) **else return** Signature Invalid
- 6) **end if**.

Ensure: valid or invalid.**Decentralized Consensus**

The IOTA 2.0 protocol works in a cycle, as presented in Figure 2, to ensure seamless interactions among its modules. This cyclical process involves messages entering the network through the congestion control algorithm, which is crucial in access management. Nodes participate in voting through the Fast Probabilistic Consensus (FPC) protocol to address conflicts and determine which branches should be discarded (IOTA, 2021). The FPC voting system is safeguarded against attackers using mana, a reputation system restricting voting privileges. Following the rejection of erroneous branches by FPC, tips are selected from the correct branches to serve as references for new messages. As these correct branches accumulate more messages, their approval weight increases. Once a branch attains sufficient approval weight, its transactions are finalized and recorded in the ledger, and the mana is updated to determine the next set of FPC voters (Drasutis, 2021).

- **New Message Layout:** IOTA 2.0 uses messages that contain data and transactions. Messages must reference 2 to 8 others, forming a dynamic DAG called the Tangle. They consist of a header, payload, and signature, maintaining system

Figure 2 Overview of the consensus applications in IOTA 2.0 protocol.

integrity. Messages end with a signature for integrity assurance, utilizing Ed25519 signatures.

- **Sybil Protection System (Mana):** DLTs use Sybil protection mechanisms like Proof of Work and Proof of Stake to prevent manipulation by attackers using multiple identities. In IOTA 2.0, this protection is called “mana.” Mana is assigned to a specific node ID by token holders during value transactions, with the amount of mana tied to the transaction’s iota quantity. Mana allocation is recorded in the ledger and serves two roles: “consensus mana” for network participation and “access mana” for network access.
- **FPC:** It is a leaderless consensus protocol where nodes interact with a limited number of others in each round to gather opinions. Consensus mana plays a crucial role by offering Sybil protection, giving more weight to trustworthy participants. Random thresholds and a decentralized Random Number Generator (dRNG) committee are used to ensure robustness and security. The dRNG committee, chosen based on mana, collaboratively creates a private key for generating random numbers. Beacon messages from committee members are required to reveal the random number, enhancing transparency and integrity.
- **Select Tips:** After FPC is completed, the protocol efficiently rejects branches that contain disliked transactions. When creating a new message, nodes randomly choose tips from the remaining branches. The tip selection algorithm involves selecting and approving between two and eight previous messages from a list of tips. This approval mechanism represents a belief in the Tangle, as it signifies the trust and confidence in the validity of the approved message and its entire history.
- **Approval Weight:** A branch’s approval weight corresponds to the total consensus mana held by nodes that have messaged on that branch. As honest nodes consistently connect their new messages to unrejected branches, these branches swiftly accumulate substantial approval weight. Conversely, rejected branches maintain minimal approval weight. For newcomers to the network, assessing the weight of branches provides insights into the outcomes of past FPC votes they may have missed.
- **Finalization:** FPC protocol ensures network consensus but has a potential for manipulation. The finality module uses trust levels from honest nodes to evaluate message irreversibility, enhancing authenticity and reducing manipulation. The Approval Weight tool relies on active consensus Mana to assess trustworthiness, considering the proportion of Mana from nodes approving the message. Messages become confirmed when they exceed a threshold, typically 70%, within 10 seconds, making them suitable for Tangle-based applications.
- **Ledger state:** To enable rapid consensus, an Unspent Transaction Output (UTXO) model is used for real-time transaction validation. This model disconnects balances from addresses, allowing users to specify which funds they want to spend, even if funds are in the same address. The “reality-

based ledger state” helps address double-spending by representing various ledger state perceptions, forming the reality ledger State. This efficient system aids nodes in understanding Tangle outcomes without exhaustive traversal.

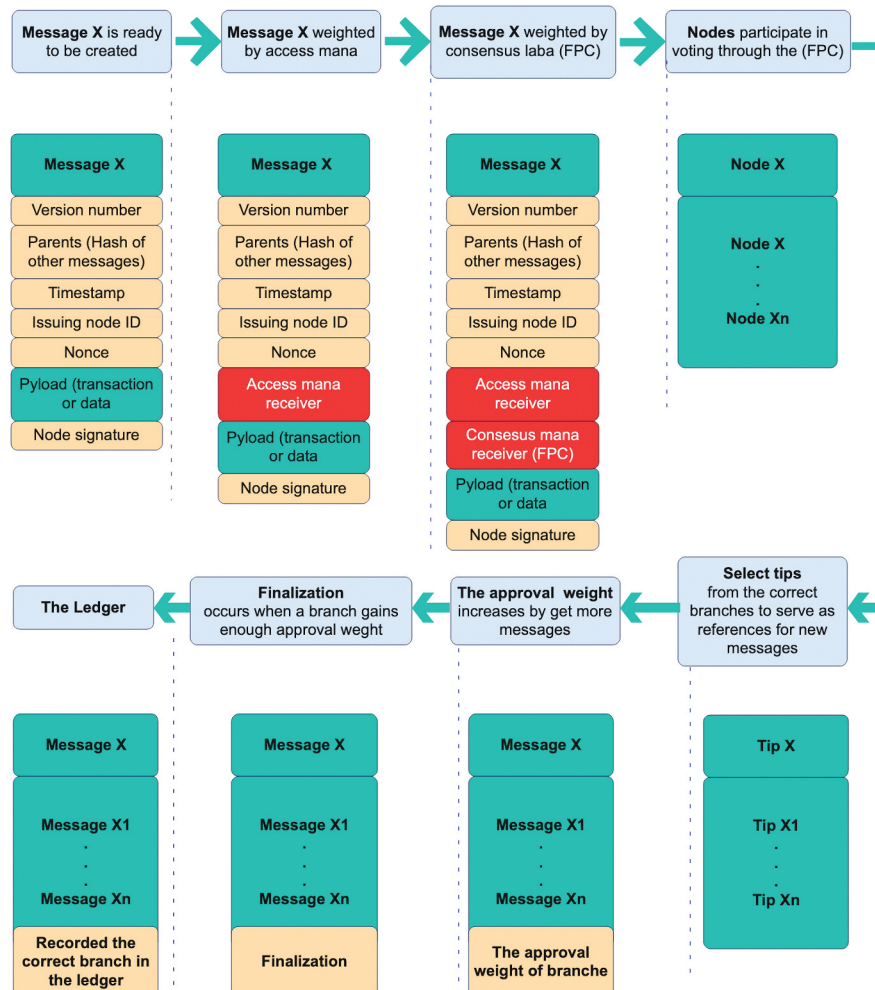
Figure 3 shows, in detail, the process of the consensus applications in the IOTA 2.0 protocol to create a new bundle.

Smart Contracts

IOTA 2.0 significantly improves decentralization, security, scalability, and consensus mechanisms. Additionally, the introduction of smart contract functionality expands the capabilities of the IOTA platform, enabling the development of decentralized applications (DApps) and opening up new avenues for utilization.

The IOTA Smart Contracts Protocol (ISCP) adopts off-chain smart contracts, meaning the execution and security of contracts

Figure 3 *Process of the consensus applications in IOTA 2.0 protocol.*



are not reliant on the entire main network (Drasutis, 2021). Instead, smart contracts operate on subchains connected to the main Tangle and are maintained by a subset of nodes known as the committee.

The committee achieves consensus and executes all contracts on its chain, thereby updating the main Tangle and, consequently, the entire network with signed transactions reflecting their states. This approach results in notably low transaction costs, making forecasting easier. Additionally, ISCP's operation does not add to the operational stress of the main network, and the security of smart contracts can be enhanced by increasing the committee's size.

The implementation of ISCP comes with several benefits and advantages. Unlike other smart contract platforms that use their own languages, ISCP supports the Ethereum virtual machine. This compatibility allows the execution of Solidity-based smart contracts from Ethereum directly on the IOTA Tangle network. Consequently, existing Ethereum contracts can be repurposed, taking advantage of the fee-less transactions and improved execution speed offered by the IOTA network.

Smart contracts in ISCP are set up and deployed by their owners, but the committee manages their execution and maintenance. Owners can offer a variable reward to incentivize committee nodes to process their contracts. The fees for the contract depend on factors such as the chain, specific contract details, and the contract owner.

IMPLEMENTATION OF THE IOTA SYSTEM

This section provides comprehensive practical examples of IOTA functionalities using Python. It covers a wide range of operations, including connecting to nodes, generating a Seed, generating addresses, getting balance, getting output (UTXO), finding output (UTXO), sending transaction (tokens), sending Data_Message, fetching Data_Message, and MQTT Protocol.

Here are a few examples; refer to the [Appendix](#) for a complete list of examples.

- 1) **Example 1:** The Client class constructor offers various parameters to enable precise process management (See [Figure 4](#)). Among the the most frequently used ones are:
 - a) *network*: Choose between devnet or mainnet to determine node selection.
 - b) *node*: Allows specifying the address of the running IOTA node in the format <https://node:port>.
 - c) *node pool urls*: Support for managing a pool of nodes, automatically selected based on their sync status.

Figure 4 Connecting to Node.

```
import iota_client
client = iota_client.Client (
    nodes_name_password=[['https://api.lb-0.h.chrysalis-devnet.iota.cafe']]
)
print(client.get_info())
```

- d) *local pow*: If set to True (default), perform Proof-of-Work locally, not remotely.
- 2) **Example 2**: The IOTA protocol utilizes an 81-trytes seed for private key generation in a single account. Additionally, it introduces a method to represent the seed as a mnemonic

Figure 5 *Generating a seed.*

```
import os
import hashlib

rnd_seed = hashlib.sha256(os.urandom(256)).hexdigest()
print(rnd_seed)
```

sentence consisting of easily understandable words, as shown in [Figure 5](#).

- 3) **Example 3**: An IOTA address is created using the *Client*. *get_addresses()* function, resulting in a list of tuples containing the generated addresses, as shown in [Figure 6](#).

Figure 6 *Generating addresses.*

```
import iota_client

client = iota_client.Client(
    nodes_name_password=[['https://api.lb-0.h.chrysalis-devnet.iota.cafe']]
)
# Insert your testing seed here.
IOTA_SEED = 'e92ed03d9d4aef97538c379c6952dae9e56d22123b0e350d1c73777da1c42de7'
address_changed_list = client.get_addresses(
    seed=IOTA_SEED,
    account_index=0,
    input_range_begin=0,
    input_range_end=10,
    get_all=True
)
print(address_changed_list)
```

- 4) **Example 4**: [Figure 7](#) showcases creating a message with a custom index and data. [Figure 8](#) fetches and prints a message with the given ID.

Figure 7 Posting Data_Message.

```
import iota_client
import unicodedata

INDEX = "IOTA network"
DATA = "Python Workshop!".encode()
client = iota_client.Client(nodes_name_password=[['https://api.lb-0.h.chrysalis-devnet.iota.cafe']])
message_id_indexation = client.message(index=INDEX, data=DATA)
print(f"Message sent!\n https://explorer.iota.org/devnet/message/{message_id_indexation['message_id']}")
print("Copy the ID of your message. You'll need it in the next example.")
print(f"Message ID: {message_id_indexation['message_id']}")
```

Message sent!
<https://explorer.iota.org/devnet/message/a4c6ec7dfc4efe91b70cb0a7e7e73dcd8b95cbb6de9b572ff83f489dc908b88>
 Copy the ID of your message. You'll need it in the next example.
 Message ID: a4c6ec7dfc4efe91b70cb0a7e7e73dcd8b95cbb6de9b572ff83f489dc908b88

Figure 8 Getting Data_Message.

```
import iota_client

MESSAGE_ID = 'a4c6ec7dfc4efe91b70cb0a7e7e73dcd8b95cbb6de9b572ff83f489dc908b88'
client = iota_client.Client(nodes_name_password=[['https://api.lb-0.h.chrysalis-devnet.iota.cafe']])
message = client.get_message_data(MESSAGE_ID)
message_index = message['payload']['indexation'][0]['index']
message_content = message['payload']['indexation'][0]['data']
print(f"Message data: {message}")
print(f"Message index: {bytes.fromhex(message_index).decode('utf-8')}")
print(f"Message content: {bytes(message_content).decode()}")
```

Message data: {'message_id': 'a4c6ec7dfc4efe91b70cb0a7e7e73dcd8b95cbb6de9b572ff83f489dc908b88', 'net
 Message index: IOTA network
 Message content: Python Workshop!

- 5) **Example 5:** Figure 9 generates a transaction with a custom index and data. The specified amount of IOTA, stored in the "amount" variable, will be transferred to the provided address.

Figure 9 Getting Data_Message.

```
import iota_client

SENDER_SEED = 'f082937eb06ea99695ad2f806ffa84128a565fb812d2e099af3a42c248eac1a8bdf486a84af3e8cfd31869a2a765fafc76ad0c1a5acf111e2c02ae8f3e88da1c'
RECIPIENT_ADDRESS = 'atoi1qp2whhc5w2ayx8y8svq7m08Suj73l27msg4yJ8lj3hjmc6ffhehshpqzac'
INDEX = "IOTA network"
DATA = "Here are your IOTA tokens!".encode()
client = iota_client.Client(nodes_name_password=[['https://api.lb-0.h.chrysalis-devnet.iota.cafe']])

output = {
    'address': RECIPIENT_ADDRESS,
    'amount': 2000000 # 2 MIOTA
}

message_id = client.message(seed=SENDER_SEED, index=INDEX, data=DATA, outputs=[output])
print(f"IOTAs sent!\n https://explorer.iota.org/devnet/message/{message_id['message_id']}")
```

CONCLUSION

The IOTA Tangle 2.0 cryptocurrency presents several attractive characteristics, including high scalability, zero-fee transactions, higher achievable transactions per second, lower energy consumption, decentralized consensus, and smart contracts. These characteristics make it well-suited for various applications, including the Internet of Things, supply chain management, and micropayments. IOTA Tangle 2.0 signs transactions using the Ed25519 scheme and achieves decentralized consensus through a leaderless FPC protocol. The IOTA Tangle 2.0 network is still under development, but it has the potential to revolutionize the way we interact with the digital world. The first fully decentralized network that supports smart contracts, IOTA Tangle 2.0 has the potential to enable new and innovative applications that were not possible before.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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APPENDIX

We shared the complete list of Python examples of IOTA functionalities at: <https://github.com/MedFartitchou/IOTA-Python>